

MEMBER 4

BI & Presentation Lead

Power BI dashboard · 12-slide deck · README

Name: _____

The shared table contract, metric definitions, risks and timeline are in **00_Team_Overview**. Read it first. This brief only covers your role.

YOUR FIRST 3 HOURS

1. **Hour 1: agree the contract with the team.** The 8 tables and the metric definitions are in 00_Team_Overview.
2. **Wireframe the 4 pages:** Executive Overview, Root Causes & Seasonality, Route Corridors, ML Risk.
3. **Install the ClickHouse ODBC driver first.** Then update Power BI Desktop to a recent version. The ClickHouse docs say the connector is on by default from version 2.137.751.0.
4. **Connect to Member 1's dummy table in DirectQuery.** ClickHouse port 8123 must be reachable from Windows. This must work by hour 3.
5. **If it fails:** check the ODBC driver, the Power BI version, and port 8123 / the VM network. Fallback: export the tables to CSV and use Import mode. Tell the team the result by hour 3.

Your deliverables

- Power BI ↔ ClickHouse test (hours 0–3)
- 4-page Power BI dashboard (.pbix) with slicers: Executive Overview, Root Causes & Seasonality, Route Corridors, ML Risk
- 12-slide deck (who owns which slide: see the list below)
- Technical README
- Two full rehearsals (10–12 min), then SUBMIT

Definition of done

- ☐ Contract and metric definitions confirmed at hour 1.
- ☐ Power BI ↔ ClickHouse connection proven **by hour 3** (DirectQuery, or the Import fallback).
- ☐ 4 pages wireframed.
- ☐ KPI cards and page layouts built from the sample CSVs of Tables 1 and 4 (hours 3–8).
- ☐ Pages 1 and 2 built (hours 8–14). Pages 3 and 4 built (hours 14–20).
- ☐ End-to-end test with the real sample tables in ClickHouse (hours 20–24).
- ☐ Power BI refreshed on the full tables. Every visual checked (hours 30–36).
- ☐ 4-page .pbix with slicers finished.
- ☐ Slides 1–6 built (hours 24–30). Slides 7–12 built with screenshots (hours 36–42).
- ☐ Technical README done. Member 1 assembles the master README.md in hours 42–48.
- ☐ Two full rehearsals done (10–12 min). **SUBMIT.**

Your schedule, hour by hour

Hours count from kickoff. Text is taken from the plan's schedule tables.

When	What you do (from the plan)
Day 1 — foundation, parallel work on the 1-month sample (hours 0–24)	
Hours 0–3	Wireframe the 4 pages. Install ClickHouse ODBC driver, update Power BI Desktop, connect to Member 1's dummy table (DirectQuery). If it fails, use the Import fallback.
Hours 3–8	Build page layouts and KPI cards from the sample CSVs of Tables 1 and 4.
Hours 8–14	Pages 1 and 2 (Executive Overview, Root Causes & Seasonality).
Hours 14–20	Pages 3 and 4 (Route Corridors, ML Risk).

When	What you do (from the plan)
Hours 20–24	End-to-end test with the real sample tables in ClickHouse (connection already proven in hours 0–3).
Day 2 — full 48-month run, polish, defense readiness (hours 24–48)	
Hours 24–30	Slides 1–6.
Hours 30–36	Refresh Power BI on full tables; check every visual.
Hours 36–42	Slides 7–12 with screenshots.
Hours 42–48	Two full rehearsals (10–12 min). SUBMIT.

Handoffs: who sends what, by when

You receive from	What	By when
Member 1	ClickHouse running, <code>flight_analytics</code> + dummy table, port 8123 reachable from Windows	Hours 0–3
Member 2	Sample CSVs of Tables 1 and 4	Hour 8
Member 2	All sample CSVs	Hour 12
Members 2, 3	Sample aggregates (Member 2) and predictions (Member 3) loaded in ClickHouse	Member 2: hours 14–20 Member 3: hours 20–24
Member 2	Final aggregates loaded into ClickHouse, row counts verified	Hours 30–36
Member 3	Confusion matrix, precision / recall, top drivers; then ROC curve and feature-importance visuals	Hours 30–42 †
Member 1	Run times and storage sizes; green Airflow tree-view screenshot	Hours 30–42 †
Members 1, 2, 3	Content for the slides each one owns	Before slides 1–6 (hours 24–30) and 7–12 (hours 36–42) †

You send to	What	By when
Team	Result of the Power BI ↔ ClickHouse test: DirectQuery works, or Import fallback	By hour 3 †
Member 1	If the connection fails: what you found (ODBC driver, Power BI version, port 8123, VM network)	Hours 0–3 †

† Not written as a handoff in the plan, but the schedule needs it. Agree the exact hour with the other member.

Slide ownership: the 12-slide deck

The member in brackets owns the content of that slide. Your schedule has you building slides 1–6 (hours 24–30) and 7–12 with screenshots (hours 36–42).

Slide	Content	Owner
1	Title, team, NTI Big Data specialization.	Member 4
2	Business problem & motivation.	Member 4
3	Dataset (BTS 2022–2025, 27.4M flights, 109 attributes, 12.5 GB).	Member 1
4	End-to-end architecture (CSV → HDFS → Parquet → Spark → ClickHouse → Power BI, Airflow on top).	Member 1
5	Storage & relational ingestion (HDFS layout + Sqoop import; state honestly if Sqoop fell back).	Member 1
6	Spark processing & cleaning rules (null handling, <code>Dep_Hour</code> , metric definitions).	Member 2
7	Core findings (Q1 worst airlines, Q2 congested airports, Q7 busiest routes).	Member 2
8	Temporal & root cause analysis (Q3, Q4, Q5, plus Q6 cancellations).	Member 2
9	ML formulation (binary classification, leak-free features, train 2022–23 / test 2024).	Member 3

Slide	Content	Owner
10	Model evaluation & feature importance (AUC-ROC, honest interpretation).	Member 3
11	Power BI dashboard showcase (4 pages).	Member 4
12	Conclusions, Airflow orchestration, future work: real-time streaming with Kafka + Spark Structured Streaming.	Member 4

Definitions you need in Power BI

Copy of the shared definitions. 00_Team_Overview is the master.

- **Delay_Rate_Pct** = Delayed / (Delayed + OnTime) × 100. **Cancellation_Rate_Pct** = Cancelled / Total_Flights × 100.
- **Dep_Hour** = `floor(CRSDepTime / 100) % 24`. **Day_Of_Week**: 1 = Monday ... 7 = Sunday (BTS `DayOfWeek`).
- **Season** (Power BI): Dec–Feb Winter, Mar–May Spring, Jun–Aug Summer, Sep–Nov Fall.
- **Route** is directed (`Origin-Dest`). **Cancellation_Reason**: A = Carrier, B = Weather, C = NAS, D = Security.

Warnings for your role

- **Test the Power BI ↔ ClickHouse connection in hours 0–3, not hour 24.** It is the riskiest integration. Use Member 1's dummy table.
- **Install the ClickHouse ODBC driver first**, then connect. Use a recent Power BI Desktop.
- **If DirectQuery fails, use Import mode.** Export the tables to CSV and load them. The dashboard works the same. Only the "DirectQuery" claim goes away, so do not claim DirectQuery in the deck if you fall back.
- **Slide 5 must be honest.** If Sqoop fell back to `hdfs dfs -put`, say so. Do not claim it ran.
- **Kafka / streaming is not built.** It appears only on slide 12 as future work.
- **Use the definitions exactly.** `Season` is computed in Power BI from the month. `Day_Of_Week`: 1 = Monday ... 7 = Sunday. Routes are directed.
- **Build on the sample CSVs first** (Tables 1 and 4 by hour 8, all by hour 12). Then switch to ClickHouse. On Day 2, refresh on the full tables.
- **If we fall behind:** Power BI page 4 (ML Risk) is cut first. **Never cut Power BI pages 1–3.**

Connection checklist (no code for your role)

- ClickHouse **ODBC driver** installed first
- Recent **Power BI Desktop** (ClickHouse connector, DirectQuery)
- Port **8123** reachable from Windows (check the VM network if not)
- Fallback: export the tables to CSV and use **Import mode**